PART A **[70 MARKS]**

QUESTION 1 **[4 Marks]**

Program 1 is the program that is used by the parking attendant to calculate the fee for parking based on the hours and rates. The parking building consists of employee and non-employee car with different type of rates:

(a) Employee of the company: RM 1.50 per hour

(b) Non-employee: RM 3.00 per hour

Based on the comments given in **Program 1**, fill in the blank lines with appropriate C++ statements. The program should generate the output as shown in **Figure 1**. *Note:* The **bold texts** in **Figure 1** indicate input from the user. (4 marks)

```
The number of cars: 2

Hours parked for car 1: 3
Hourly pay rate for car 1: 1.50

Hours parked for car 2: 5
Hourly pay rate for car 2: 3.00

Here is the total fee for each car:
Car [1] = RM4.50
Car [2] = RM15.00
```

Figure 1: Example runs of the Program 1

```
1   // Program 1
2   #include<iostream>
3   #include<iomanip>
4   using namespace std;
5
6   int main()
7   {
8     const int no_cars = 2; //Constant for the array size
9
10    //Declare an array named hours and rate to allocate 2
11    //elements of type integer and double respectively.
12    //Use constant declared in line 8.
```

```
13      _____________________________________________________________
14      _____________________________________________________________
15
16    cout << "The number of cars: " << no_cars;
17    cout << endl;
18
19    for (int index = 0; index < no_cars; index++)
20    {
21      cout << "\nHours parked for car " << (index+1) << ": ";
22      //Read a value and assign it into array named hours
23      cin >>_____________________________________________________
24      cout << "Hourly pay rate for car " << (index+1) << ": ";
25      //Read a value and assign it into array named rate
26      cin >>_____________________________________________________
27    }
28    cout << "\nHere is the total fee for each car:\n";
29    cout << fixed << showpoint << setprecision(2);
30
31    for (int index = 0; index < no_cars; index++)
32    {
33       //Calculate a value of totalPay (fee for parking)
34       double totalPay =_____________________________________________
35       cout << "Car [" << (index + 1);
36       cout << "] = RM" << totalPay << endl;
37    }
38     return 0;
    }
```

QUESTION 2 **[16 Marks]**

Given the declaration and initialization of some parallel arrays as in the following code segment:

```
string patients[] = {"Wendy", "Kumar", "Ros", "Mael"};
int age[] = {25, 43, 32, 54};
double averageSugarLevel[4];
```

Based on the concept of arrays and using loops, answer the following questions:

(a) Define a two-dimensional array named **`sugarLevel`** which has large enough space to hold the data in the following table. Initialize the array with values from the table. (2 marks)

4.5	5.7	6.8
6.2	7.1	7.3
5.1	6.3	8.0
7.5	8.5	7.3

Answer:

(b) Write **a code segment** that prints the patient names along with their ages. The output should be as follows: (3 marks)

```
Age of Patients
---------------
Wendy     25
Kumar     43
Ros       32
Mael      54
```

Answer:

(c) Write **a code segment** that produces the following output from the array named **`sugarLevel`** that you defined in (a). (4 marks)

```
Name          Breakfast   Lunch   Dinner
Wendy           4.5         5.7     6.8
Kumar           6.2         7.1     7.3
Rose            5.1         6.3     8.0
Mael            7.5         8.5     7.3
```

Answer:

(d) Write a **code segment** to calculate the average sugar level of the patient. The code also store the average sugar level value in the array **`averageSugarLevel`** and print the list of the patients' average sugar level value as follows. (7 marks)

```
Sugar Level of Patients
-----------------------
Wendy     25
Kumar     43
Ros       32
Mael      54
```

Answer:

QUESTION 3 **[11 Marks]**

(a) Name at least two member functions associated with **`iostream`** object. (2 marks)

Answer:

(b) Suppose two file stream objects named **`fin`** and **`fout`** have been declared in the code as follows:

```
ifstream fin;
ofstream fout;
```

Complete the code so that the program can copy 10 numbers from a file named **`file1.dat`** and save the numbers to another file named **`file2.dat`** (copy 10 numbers between files). The program should also include an error checking for opening the files. (9 marks)

Answer:

QUESTION 4 **[4 Marks]**

Based on the **Program 2**, write the output into **Table 1**. *Note:* one cell represents one space per digit or character. (4 marks)

```
1   // Program 2
2   //This program write three rows of numbers to a file
3   #include <iostream>
4   #include <fstream>
5   #include <iomanip>
6   using namespace std;
7
8   int main ()
9   {
10      const int ROWS = 3;
11      const int COLS = 3;
12      double nums [ROWS][COLS] = {1501, 3.579, 984,
13                                  5.3, 91, 1979,
14                                  24, 606, 11.0287};
15      fstream outFile ("Jadual.txt", ios::out);
16
17      for (int row = 0; row < ROWS; row++)
18      {
19          for (int col = 0; col < COLS; col++)
```

```
20          {
21              outFile << setprecision (1) << fixed << showpoint;
22              outFile << setw(6) << nums [row][col];
23          }
24          outFile << endl;
25      }
26      outFile.close();
27      return 0;
28  }
```

Table 1: Output file named **`Jadual.txt`**

QUESTION 5 **[4 Marks]**

As you are travelling to the United States, it is important to know the temperature for you to be able to prepare what to wear for the day. Apparently in the United States, they use Fahrenheit and not Celsius. A conversion program that will ask for a temperature in Fahrenheit, convert that value to Celsius, and display the result, would definitely become handy. Complete the **Program 3**. You need to write the function **`convert`**. *Note:* $C = 5 / 9 \times (F - 32)$. Do not modify the function **`main`**. (4 marks)

```
1   // Program 3
2   #include <iostream>
3   #include <iomanip>
4   using namespace std;
5
6   //Write your function prototype here
7   _____________________________________________________________
8   int main ()
9   {
10      double temperature;
```

```
11
12      cout << "Enter a temperature in Fahrenheit, and I will
13           << convert it to Celsius: ";
14      cin >> temperature;
15      convert (&temperature);
16      cout << fixed << setprecision(4);
17      cout << "Value in Celsius: " << temperature << endl;
18
19      return 0;
20  }
21
22  //Write the function convert here
23  _____________________________________________________________
24  _____________________________________________________________
25  _____________________________________________________________
26  _____________________________________________________________
27  _____________________________________________________________
```

QUESTION 6 **[10 Marks]**

(a) Determine the output and draw a memory layout (or memory allocation) of the pointers and variables for code segment below: (7 marks)

Note: Draw a memory layout that represents C++ statement line by line.

```
int  x = 10, y = 20, z = 30;
int *ptr;

cout << x << " "  << y << " " << z << endl;
ptr = &x;
*ptr *= 10;
ptr = &y;
*ptr *= 4;
ptr = &z;
*ptr *= 2;

cout << x << " "  << y << " " << z << endl;
```

Answer:

(b) Determine the output for code segment below: (3 marks)

```
int numbers[5] = {2,4,6,8,10};
int *numPtr;

numPtr = numbers;
cout << *(numbers + 3) << endl;
cout << *(numbers) << endl;

for (int count=0; count < 2; count++)
{
    cout << *numPtr  << "\t";
    numPtr++;
}
cout << endl;

for (int count=0; count < 2; count++)
{
    numPtr--;
    cout << *numPtr << "\t";
}
```

Answer:

QUESTION 7 **[5 Marks]**

The code segment below has five (5) syntax and/ or logical errors. Identify and describe each of the errors. (5 marks)

```
struct TwoLoc
{
   int x = 10;
   int y = 15;
}

int main()
{
   TwoLoc point[10];
   point.x[0] = 1;
   point.y[0] = 1;
   cout << point << endl;
   return 0;
}
```

Answer:

QUESTION 8 **[16 Marks]**

Meteorological Department is responsible to keep the information about weather forecast in Malaysia and display the information through the website. Write the code segment for each of the following tasks:

(a) i) Declare a structure type named **`TempScale`**, with the following members:

`fahrenheit` : a double value

`celsius` : a double value

ii) Declare a structure type named **`Reading`**, with the following members:

`windSpeed` : an integer value

`humidity` : a double value

`temperature`: a **`TempScale`** structure variable

iii) Declare a variable of structure type **`Reading`** named **`today`**. (6 marks)

Answer:

(b) Write statements that will store the following data into the variable you declared in (a).

Wind speed: 8 km/h
Humidity: 53%
Fahrenheit temperature: 80.2 degrees
Celsius temperature: 26.8 degrees

(3 marks)

Answer:

(c) Define a function named **`showReading`**. It should accept a **`Reading`** structure variable as its argument. The function should display the contents of the variable onto the screen. (7 marks)

Answer: